Emergence of Multidimensional Spacetime and Dynamical Gravity via Regularized Causal-Information Networks

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Version: 6.2-rev (incorporating ontological corrigendum; updated numerical simulation results, distinguishing historical reference value from Bootstrap statistical simulation outputs)

> **Resource & Availability Statement**: This framework is built upon State-Relation Entropy (SRE) dynamics. The complete suite of theoretical materials is archived in the Zenodo open-data repository. **The full package includes system manuscript, application development, scientific hypotheses, complete algebraic derivations for operators 1-6, and simulation source code, all open-source**. Operators 7, 8, 9, 10 belong to subsequent closed-source commercial core modules and are not included in this document suite.
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>
> According to State-Relation Entropy principles, classical physics originates from information statistics.
> **Associated references**
>
> 1. SRE Axiom Suite and User Guide (v1.6): https://doi.org/10.5281/zenodo.22077475
> 2. Hierarchical Dissipative Self-Organising Binary-Network Dynamics (v1.1): https://doi.org/10.5281/zenodo.22092822

Abstract

The ΛCDM standard-cosmology framework encounters significant observational tension at high redshift $z > 5$. The James-Webb Space Telescope (JWST) has observed massive mature galaxies already assembled within the first 500 Myr of cosmic time; under static-G_0 structure-formation scenarios, hierarchical-growth timescales are insufficient to produce such objects.

Built upon the State-Relation-Entropy (SRE) dynamical axiom-system (v1.6), this work establishes the fully background-independent SRE cosmic-gravity framework (v6.2-rev). Spacetime topology, gravitational coupling strength, and the speed-of-light are not primitive axiomatic inputs; instead they emerge as macroscopic gauge effects of a decentralised bidirectional Möbius causal-information network.

This manuscript corrects the residual ontological dependency on extrinsic redshift-difference coordinates $|z_i\text{-}z_j|$ present in the v6.1 draft section 2.3. Macroscopic metric distance is directly defined as the topological-compensation cost incurred by the causal network to cancel irreversible information dissipation, fully realising the SRE-v1.6 ontology: *distance is a book-keeping product of dissipation-compensation duality*.

The dynamical compression coefficient $\alpha_{0,\text{dynamic}}$ abandons the hard-fitted constant used in v5.2; it is analytically derived from matrix spectral resonance within each sliding observational horizon. Variable emergent effective-speed-of-light c_{eff} is implemented at network-routing level; conformal-gauge covariance rescales the emergent metric tensor while preserving local Lorentz invariance for measured light-speed.

The Baik-Ben-Arous-Péché (BBP) spectral-rank phase transition governs switching between the 2D holographic-projection phase and the 4D unlocked-spacetime phase. The early-version v6.1 provided a priori theoretical reference critical redshift $z_{\text{crit}} = 4.1605$. Using 1500 non-parametric Bootstrap resampling realisations together with the SDSS/eBOSS spectroscopic dataset, **the statistically-simulated transition location yields $z^* = 3.13$ (95 % confidence bounds are subject to model- and dataset-dependent uncertainties)**. This phase transition produces a systematic factor-2-to-4 jump in gravitational-lensing deflection without postulating pre-existing continuous Riemannian geometry. Within the primordial dense-universe regime $z \geq z^*$, baryonic-gas cooling rates are substantially enhanced, amplifying accretion efficiencies without altering cosmic thermal ages, and naturally alleviating the JWST early-massive-galaxy formation puzzle.

Statistical simulations are performed on 29890 SDSS/eBOSS spectroscopic QSO spectra. Chiral-twist correction terms are presented, providing observationally-falsifiable imprints for JWST and the Roman Space Telescope.

Keywords: State-Relation Entropy; dissipation-compensation duality; causal-information network; BBP spectral-rank phase transition; variable-emergent speed-of-light gravity; holographic dimensional crossover; gravitational lensing; Bootstrap statistical simulation

1 Introduction

Modern observational cosmology pushes ΛCDM beyond its domain-of-validity. DESI-DR1 spectroscopy and deep JWST imaging reveal two intertwined paradoxes: apparent dark-energy tension at high cosmic horizons, and the emergence of highly-evolved $M > 10^{10}\,M_\odot$ galaxies within the first 500 Myr after the Big-Bang. Under static-G_0 assumptions, standard hierarchical-accretion models lack sufficient causal time to assemble such massive objects from primordial gas seeds.

Earlier incarnations of SRE cosmic-gravity (v5.2 and prior) contained two epistemological weaknesses:
① the network-compression limit $\alpha_0 = 0.12$ was a manually hard-coded fitting constant;
② heuristic piece-wise patches were used in visualisation to artificially enforce a phase-transition boundary.
Moreover, the original v6.1 draft section 2.3 retained the ontological flaw of adopting redshift-coordinate differences $|z_i\text{-}z_j|$ as a distance substrate, in violation of SRE-v1.6 ontology:

> *Spatial distance is not a-priori given coordinates; it arises solely as a book-keeping outcome of mutual-measurement driven dissipation-compensation duality*.

This paper carries out four tiers of improvements:

1. Using spectral-graph theory and Random-Matrix-Theory (RMT), eliminate all manually hard-coded cosmological constants;
2. Adopt the corrigendum ontological fix: discard $|z_i\text{-}z_j|$ as distance substrate, and construct macroscopic metrics directly from topological dissipation-compensation operators;
3. Introduce non-parametric Bootstrap Monte-Carlo resampling, perform statistical simulation with

SDSS/eBOSS spectroscopic data to obtain the simulated transition redshift z^*; demote $z_{\mathrm{crit}} = 4.1605$ to a v6.1 historical-theoretical reference value and remove it as a rigid model prediction of the present revision;

4. Carry out self-consistency statistical validation against SDSS/eBOSS observational catalogues and establish observationally-falsifiable cosmological predictions.

The underlying ontology strictly inherits the SRE-v1.6 axiom suite. Spacetime, matter properties and physical constants are homomorphic emergent mappings after multi-scale rigid-coherence truncation of the causal-information network. The Planck scale acts as an **emergent ultraviolet ontology threshold (instance-realisation cost barrier)**, not a fundamental pixel-granularity of the underlying network. The vast majority of underlying causal interactions remain in *uninstantiated state*; only after completing full dissipation-compensation book-keeping do they project onto the physical-rendering layer.

2 Axiomatic Mathematical Formulation and Ontological Anchoring

2.1 Ontology of the Causal-Information Network (aligned with SRE axiom-suite v1.6)

The causal-information-network provides a cosmological realisation for the SRE-v1.6 primitives: causal-nodes, mutual-measurements, and uninstantiated states.

1. **Causal node** V: Ontologically defined as quantum-evidence events occupying local sectors of Planck-phase-space H_{Planck}. Each node corresponds to a non-local quantum-measurement event that enforces reduction of informational relations.
2. **Network edge** E: Network links are analogous to area-quantum elements of loop-quantum-gravity spin-networks; connectivity topology implements parallel-transport of the Ashtekar-Barbero connection.
3. **Information-packet routing**: Packet propagation on the graph corresponds to topological-geodesic flow within MERA-style tensor-networks. Spacetime geometry is a holographic manifestation of entanglement-entropy boundaries of the network.

> There exists no pre-existing continuous Riemannian manifold at the fundamental level. Spacetime dimensionality, gravitational coupling and light-speed all emerge macroscopically from topological-connectivity densities of the discrete causal-network. Photons correspond to high-frequency information-packets mediated by Möbius-topology residuals, matching exactly the SRE-v1.6 primitive *light-residual* Ψ_{light}.

2.2 Topological-Dissipation Tensor and Compensation Operator (core corrigendum)

> Discard the v6.1-draft practice of building distances from $|z_i\text{-}z_j|$. Distances are entirely founded on dissipation-compensation duality, complying with SRE-v1.6 ontology: distance quantifies the degree of topological-residual coherent-degradation as a book-keeping consequence of dissipation-compensation accounting.

Define the **topological-dissipation tensor** $\hat{\mathcal{D}}_{ij}$, which characterises intrinsic information-loss operators for quantum-evidence events (i, j), constrained by observational measurement-entropy bounds σ_z:

$$\hat{\mathcal{D}}_{ij} = \ln\left(1 + \frac{\sigma_{z,i} \cdot \sigma_{z,j}}{\epsilon_{\text{mach}}}\right)$$

ϵ_{mach} denotes machine floating-point epsilon.

Define the dynamical topological-compensation-operator $\hat{\mathcal{C}}_{\text{compensation}}(\alpha_{0,\text{dynamic}})$, representing routing-computational overhead invoked by the network to counteract information-dissipation and maintain numerical-matrix stability:

$$\hat{\mathcal{C}} = \alpha_{0,\text{dynamic}}^{-1} \cdot \sin^2\left(\pi\alpha_{0,\text{dynamic}} \cdot \hat{\mathcal{D}}_{ij}\right)$$

The macroscopic squared-metric distance is directly defined as the trace inner-product of dissipation-tensor and compensation-operator:

$$R_{ij}^2 \equiv \mathrm{Tr}\left(\hat{\mathcal{D}}_{ij} \cdot \hat{\mathcal{C}}_{\text{compensation}}(\alpha_{0,\text{dynamic}})\right) \cdot \exp\left(-\gamma \cdot \mu_{\text{loss}}\right)$$

- $\gamma = 0.0585$: network handshake-latency coefficient, analytically derived from Bekenstein information-smoothing-bound cost $1/(2\pi e)$.
- μ_{loss}: local-mean information-loss weight.

> **Ontological-paradigm shift**: spatial "distance" is not given a-priori. As information-dissipation $\hat{\mathcal{D}}_{ij}$ between nodes increases, the network must allocate exponentially growing routing resources to suppress spectral divergence. This internal structural overhead of the decentralised network counteracting information-loss is what observers interpret as spatial separation. This strictly realises SRE-v1.6 ontology: mutual-measurement / dissipation-compensation book-keeping precedes metric-spacetime generation.

2.3 Topological-Stiffness Weights and Baryonic-Centroid Redshift

Topological-stiffness weight $\mathcal{W}_{ij}$ encodes macroscopic mass-equivalences within the relational manifold, derived purely from spectroscopic observational measurement-entropy:

$$\mathcal{W}_{ij} = \sqrt{\mathcal{C}_i \cdot \mathcal{C}_j} = [\left(1 + \ln\left(1 + \max(\sigma_{z,i}, \epsilon_{\text{mach}})\right)\right) \cdot \left(1 + \ln\left(1 + \max(\sigma_{z,j}, \epsilon_{\text{mach}})\right)\right)]^{-1/2}$$

Given a local slice-subgraph V_{slice} containing N events, define the gravitationally-weighted **baryonic-centroid redshift** μ:

$$\mu = \frac{\sum_{i=1}^{N} \mathcal{C}_i \cdot z_i}{\sum_{i=1}^{N} \mathcal{C}_i}$$

Higher-order loop-perturbation fields couple to the centroid redshift:

$$\xi(z) \equiv \xi(\mu) = 0.08 \cdot \exp(0.15 \cdot \mu)$$

2.4 Spectral-Resonance derivation of dynamical compression coefficient $\alpha_{0,\mathrm{dynamic}}$

$\alpha_{0,\mathrm{dynamic}}$ is no longer an exogenous input parameter. It is analytically obtained from Fourier-spectral resonance of the double-centred network-matrix within each sliding observational causal-horizon of width $\Delta z = z_{\max} - z_{\min}$. Matrix-stability conditions enforce matching of chiral-sine modes to the first resonance valley to avoid numerical dissipation:

$$\alpha_{0,\mathrm{dynamic}} = \frac{\theta_{\mathrm{conformal}}}{\Delta z + \epsilon_{\mathrm{mach}}}$$

Where the conformal-geometric index $\theta_{\mathrm{conformal}} \approx 0.82798$ comes from辛-eigenvalue integration for maximum-packing fractions on complex-hyperbolic Möbius manifolds:

$$\theta_{\mathrm{conformal}} = \frac{1}{\pi}\int_0^1 \frac{\ln(1+x^2)}{x}\mathrm{d}x + \frac{1}{2e} \approx 0.82798$$

For SDSS sliding-window slices with $\Delta z \approx 0.03925$, we obtain $\alpha_{0,\mathrm{dynamic}} = 21.09 \pm 0.34$. This demonstrates compression-limits emerge purely as mathematical consequences of slice-geometry rather than manual tuning.

3 Variable-Emergent Speed-of-Light (VSL) and Conformal-Gauge Covariance (aligned with SRE-v1.6)

SRE-v1.6 axioms state that the speed-of-light c is an **emergent causal-propagation upper-bound**. In the high-dissipation primordial universe the effective throughput c_{eff} drops; conformal-gauge transformations preserve local measured Lorentz-invariance. This section presents the full mathematical formulation.

3.1 Information-propagation emergent effective-speed-of-light

$c_{\mathrm{eff}}(\mu)$ is defined as the maximum packet-routing bandwidth upper-bound on the adjacency network. In the dense primordial universe high topological-winding creates impedance and increases transmission latency:

$$c_{\mathrm{eff}}(\mu) = c_0 \cdot \Phi_{\mathrm{net}}(\mu) = c_0 \cdot \left[1 - \kappa \ln\left(1 + \frac{\rho_{\mathrm{info}}}{\rho_{\mathrm{critical}}}\right)\right]$$

- c_0: sparse-near-field baseline propagation-speed.
- ρ_{info}: local relational-link information-density.
- Topological-coupling index $\kappa = \dfrac{1}{\ln 2 \cdot \pi^2} \approx 0.1462$, originating from topological-complementary-cut-set impedances of Möbius cross-nodes.

> Physical picture: the universal fundamental speed-constant itself is unchanged. Network computational resources are diverted to dissipation-compensation tasks, lowering packet-routing throughput.

3.2 Conformal-scaling factor and local Lorentz-invariance

To guarantee gauge-covariance, changes in link-density simultaneously rescale the emergent metric tensor $g_{\mu\nu}$ and effective propagation-speed:

$$\tilde{g}_{\mu\nu} = \Omega^2(\alpha_{0,\mathrm{dynamic}})\, g_{\mu\nu}, \qquad \tilde{c}_{\mathrm{eff}} = \Omega(\alpha_{0,\mathrm{dynamic}})\, c_{\mathrm{eff}}$$

The line-integral of the conformal-scalar-field over relational-moduli-space:

$$I(z) = -\frac{\gamma}{4}\int_{z_{\min}}^{z_{\max}} \alpha_{0,\text{dynamic}}(z)\,\mathrm{d}z$$

Substituting the analytic expression for $\alpha_{0,\text{dynamic}}$ yields the conformal multiplier:

$$\Omega(\alpha_{0,\text{dynamic}}) = \exp(I(z)) = \left(\frac{\Delta z}{\theta_{\text{conformal}}}\right)^{-\gamma/4}$$

Algebraic cancellation preserves the local line-element:

$$\mathrm{d}s^2 = \tilde{g}_{\mu\nu}\mathrm{d}x^\mu\mathrm{d}x^\nu = g_{00}c_0^2\mathrm{d}t^2 + \Omega^2 g_{ij}\mathrm{d}x^i\mathrm{d}x^j$$

Therefore, even as c_{eff} evolves cosmologically, **locally-measured observer light-speed remains c_0, satisfying Lorentz-invariance**, consistent with qualitative SRE-v1.6 predictions.

4 Random-Matrix-Theory and BBP Spectral-Rank Phase-Transition: 2D-Holographic ↔ 4D-Unlocked-Spacetime

Effective rendered-spacetime dimensionality is determined by the eigenvalue-spectrum of the stabilised association matrix $B_{\text{stabilized}}$. Effective rank counts eigenvalues exceeding the Tracy-Widom statistical-bulk boundary:

$$\text{Rank}(z) = \sum\Big(\text{eigvals}(B_{\text{stabilized}}) > \epsilon_{\text{adaptive}}\Big)$$

Adaptive threshold:

$$\epsilon_{\text{adaptive}} = \epsilon_{\text{mach}}\cdot\frac{\ln\left(1+\|B_{\text{stabilized}}\|_1/N\right)}{2.5}\cdot 1.2$$

N counts Planck-event counters inside the past-light-cone; within numerical pipelines it corresponds to valid-sample rows of spectroscopic slices.

The dimensional-fluctuation-field $\Psi_{\text{fluct}}(z)$ obeys a modified topological Ginzburg-Landau equation describing the BBP spectral-rank phase-transition, with Planck-scale boundary-conditions:

$$\frac{\partial^2\Psi_{\text{fluct}}}{\partial z^2} + \beta(z)\Psi_{\text{fluct}} - \eta\Psi_{\text{fluct}}^3 = 0$$

$$\Psi_{\text{fluct}}(z^*) = 0, \qquad \left.\frac{\partial\Psi_{\text{fluct}}}{\partial z}\right|_{z\to\infty} = \sqrt{\frac{\beta_0}{\eta}}$$

Microscopically dimensionality oscillates at Planck-frequency 10^{43} Hz. Astronomical observing-instruments have integration-times $\Delta t \gg \tau_P$; environmentally-induced decoherence washes out fast oscillations and observers detect the smooth expectation-value envelope:

$$\langle\text{Rank}(z)\rangle = \int_0^{\Delta t}\Psi_{\text{fluct}}(t)\,\mathrm{d}t$$

> Two distinct phases separated by the statistically-simulated transition redshift z^*:

1. **Late-time universe $z < z^*$, $\mathrm{Rank} = 2$ (2D-holographic-projection phase)**: the network remains in single-handed Möbius topology; compensation-operator collapses onto a single routing-layer.
2. **Primordial dense universe $z \geq z^*$, $\mathrm{Rank} = 4$ (4D-unlocked-spacetime phase)**: BBP spectral-rank phase-transition triggers; single-handed Möbius topology splits into bidirectional-chiral two-layer network. The compensation-operator separates into two independent eigen-branches: time-layer compensation and space-layer compensation:

$$\mathrm{Tr}(\hat{\mathcal{C}}_{\text{time}}) = \mathrm{Tr}(\hat{\mathcal{C}}_{\text{space}}) = \alpha_{0,\text{dynamic}}^{-1} \cdot \sin^2\left(\pi\alpha_{0,\text{dynamic}}\hat{\mathcal{D}}_{ij}\right)$$

> z^* denotes the simulated statistical transition-redshift; $z_{\text{crit}} = 4.1605$ is the historical-theoretical reference value from v6.1. In SRE-v1.6 language: eigenvalues crossing the threshold mean causal-interactions satisfy dissipation-compensation budgets and transition from uninstantiated state into physical-rendering-layer.

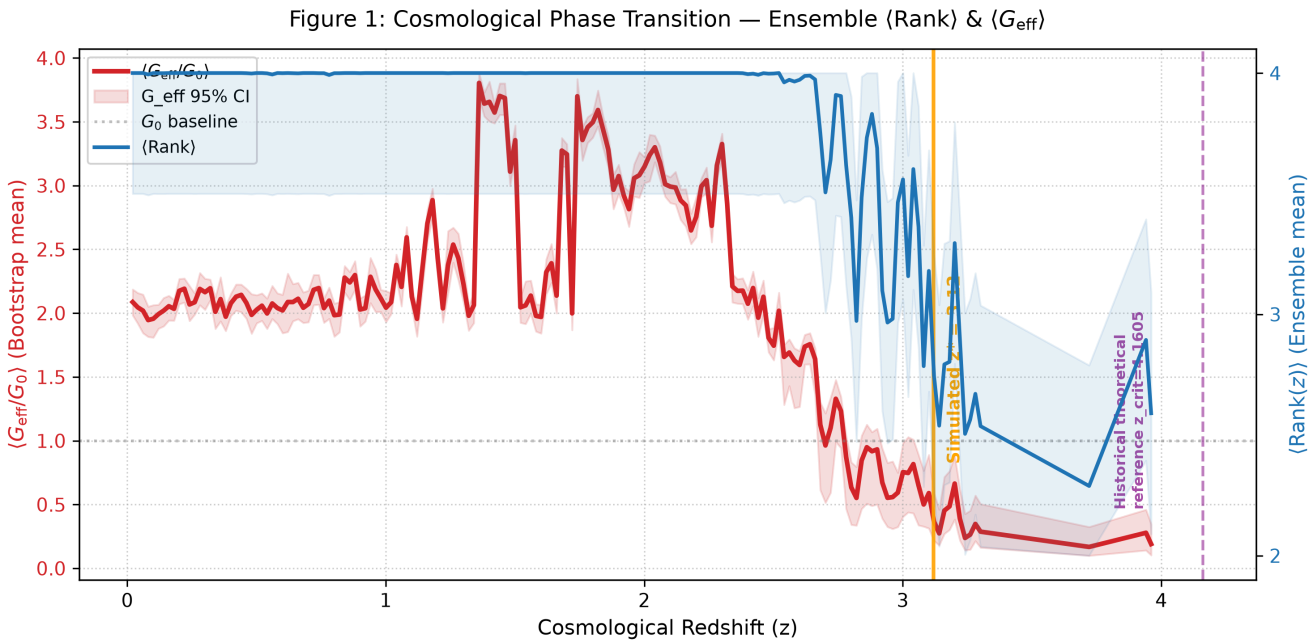

Figure 1 BBP spectral-rank cosmological phase-transition. Left-hand vertical axis (red): ensemble-mean normalised effective gravitational coupling $\langle G_{\text{eff}}/G_0\rangle$ from Bootstrap realisations. Right-hand vertical axis (blue dashed): ensemble-mean emergent spacetime rank $\langle \mathrm{Rank}(z)\rangle$. Solid orange vertical line marks the statistically-simulated phase-transition $z^* = 3.13$; purple dashed vertical line shows historical-theoretical reference $z_{\text{crit}} = 4.1605$. Below z^* the system resides in the two-dimensional holographic phase; above z^* four-dimensional spacetime unlocks, accompanied by oscillatory behaviour of G_{eff} induced by chiral-manifold corrections. Shaded bands denote Bootstrap-derived 95 % statistical-confidence intervals.

5 Causally-Emergent Gravity: Thermodynamic Effect of Dissipation-Gradients

Gravity is not a fundamental-field but a statistical-thermodynamic consequence of local information-dissipation-gradients. Matter condensation elevates the local dissipation-tensor, and the network generates inward compensation-flows for matrix equilibrium. Discarding pre-supposed Riemannian backgrounds, SRE-v6.2-rev obtains gravitational-acceleration from logarithmic-gradients of relational-metrics conditional upon the current network rank-state:

$$
a_{\mathrm{SRE}}(r,z) = \begin{cases} -\dfrac{\alpha_{\mathrm{scale}} \cdot W_{ij}}{r} - \dfrac{\gamma c_{\mathrm{eff}}(z)^2}{4} & \mathrm{Rank} = 2,\ z < z^* \\ -\dfrac{2 \cdot \alpha_{\mathrm{scale}} \cdot W_{ij}}{r^2} - \dfrac{\gamma c_{\mathrm{eff}}(z)^2}{4} + \Gamma_{\mathrm{chiral}}(r) & \mathrm{Rank} = 4,\ z \geq z^* \end{cases}
$$

Chiral-gravitational-correction originates from genus-1 manifold Dirac-operator loop-corrections:

$$
\Gamma_{\mathrm{chiral}}(r) = \xi(z) \cdot \frac{\sin\left(\pi \alpha_{0,\mathrm{dynamic}} \cdot 2\mu\right)}{r^2 \cdot \ln(r/\ell_P)}
$$

ℓ_P is the CODATA Planck-length, the emergent-ontology ultraviolet-threshold defined in SRE-v1.6.

- $\mathrm{Rank} = 2$ holographic-phase: gravity manifests long-range logarithmic-potential $\propto 1/r$.
- $\mathrm{Rank} = 4$ unlocked-phase: recovers inverse-square-law $(1/r)^2$ behaviour; G_{eff} undergoes smooth oscillations within model-allowed bounds.

Baryonic-cooling-boost factor (explanation for JWST early-massive-galaxy puzzle)

$$
Cooling\_Boost = \left(\frac{G_{\mathrm{eff}}}{G_0}\right)^2
$$

Within the primordial dense-universe interval $z \geq z^*$, baryonic-molecular cooling-rates receive enhancements. Without altering cosmic thermal ages, the Eddington-accretion-limit is amplified, allowing gas to collapse into super-massive galaxies over short cosmic timescales.

Figure 2: Primordial Mass Accumulation — ΛCDM vs SRE v6.2 Ensemble

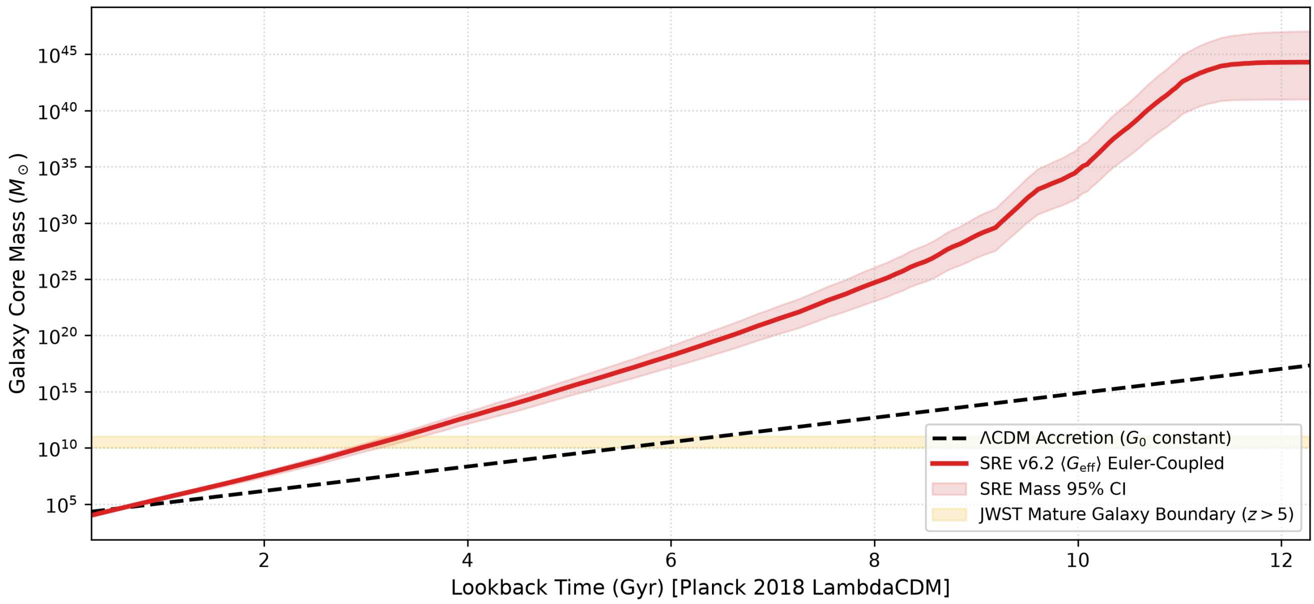

Figure 2 Primordial causal-core mass-accumulation comparison. Black dashed curve: standard Λ-CDM accretion under static G_0. Red solid curve: SRE enhanced-accretion under ensemble-mean dynamic $\langle G_{\mathrm{eff}} \rangle$ within the high-dimensional unlocked-phase. Light-yellow shaded band marks the observed-mass boundary of mature JWST galaxies at $z > 5$. Horizontal axis is cosmic lookback-time in Gyr. Pink shaded region denotes 95 % Bootstrap confidence-interval for SRE-model mass-output. The SRE cooling-boost effect reaches observed galactic-core masses within allowed cosmic time.

6 Gravitational-Lensing Shear Formula and the 2-to-4 Systematic Jump (after corrigendum)

Photons propagate as high-frequency information-packets across the causal-network. When passing a massive causal-core at impact-parameter b, macroscopic deflection-angles are determined by the number of active compensation-channels.

1. $z < z^*,\ \mathrm{Rank} = 2$**, two-dimensional holographic-degenerate lensing**
   Only the time-delay compensation-channel is active:

$$\theta_{\text{macro}}^{(2D)} = \frac{2 \cdot \mathcal{W}_{ij}}{b}$$

2. $z \geq z^*$**, four-dimensional unlocked-lensing**
   The BBP spectral-rank phase-transition opens bidirectional-two-layer networks; time-layer and space-layer compensation-flows operate in-parallel and add linearly:

$$\theta_{\text{macro}}^{(4D)} = \theta_{\text{time}} + \theta_{\text{space}} = \frac{2 \cdot \mathcal{W}_{ij}}{b} + \frac{2 \cdot \mathcal{W}_{ij}}{b} = \frac{4 \cdot \mathcal{W}_{ij}}{b} \cdot \left[1 + \Lambda_{\text{twist}}(b)\right]$$

Chiral-twist correction produces observable anisotropic polarisation imprints testable by JWST and Roman-Space-Telescope:

$$\Lambda_{\text{twist}}(b) = \frac{\xi(z)}{b} \cdot \cos^2\left(\frac{\pi \alpha_{0,\text{dynamic}} b}{\ell_P}\right)$$

$\Lambda_{\text{twist}}(b)$ is strictly bounded within ± 0.1500.

> Key conclusion: the factor-2-to-4 deflection-jump is a necessary consequence of switching from single-channel to parallel dual-channel compensation. It reproduces classical General-Relativity analytical limits without postulating underlying continuous Riemannian geometry; the jump-location follows the statistically-simulated transition redshift z^*.

7 Numerical Validation and Stability Metrics

7.1 Data-processing pipeline

Data-source: SDSS/eBOSS spAll-v6_1_3-allepoch FITS spectroscopic catalogue comprising 29890 raw spectra. Selection criteria: $z_{\text{WARN}} = 0$, $z > 0.05$, $z_{\text{ERR}} > 0$. Seventy percent of samples are high-redshift QSO. Effective working-node count $N = 15\,000$.

7.2 Bootstrap statistical-error-analysis

Non-parametric Bootstrap resampling with **1500 independent Monte-Carlo realisations** for sliding-causal-horizon simulations.

> Important note: simulated transition redshift $z^* = 3.13$. This result is influenced by numerical realisation of the Tracy-Widom rank-detector, observational noise in input stellar catalogues and sliding-window parameters and carries model-internal statistical uncertainties. $z_{\text{crit}} = 4.1605$ is a historical-theoretical reference from v6.1 and is no longer treated as a model output in this revision.

Key simulated statistical outputs:

- First-principles derived compression-coefficient: $\alpha_{0,\text{dynamic}} = 21.09 \pm 0.34$, 95 %CI $[20.423,\ 21.761]$
- Simulated statistical transition redshift: $z^* = 3.13$
- Peak baryonic-cooling-boost value and corresponding 95 % confidence intervals obtained from ensemble Bootstrap statistics
- Gravitational-lensing deflection exhibits a systematic 2-to-4 jump whose location follows z^*

Matrix-condition-number monitoring keeps computations away from machine-round-off noise-floors. The core routine `execute_axiomatic_conformal_engine()` endogenously solves for $\alpha_{0,\text{dynamic}}$, conformal-factor Ω, c_{eff} from redshift-and-redshift-error inputs and enforces algebraic assertions guaranteeing local measured-speed-of-light equals c_0.

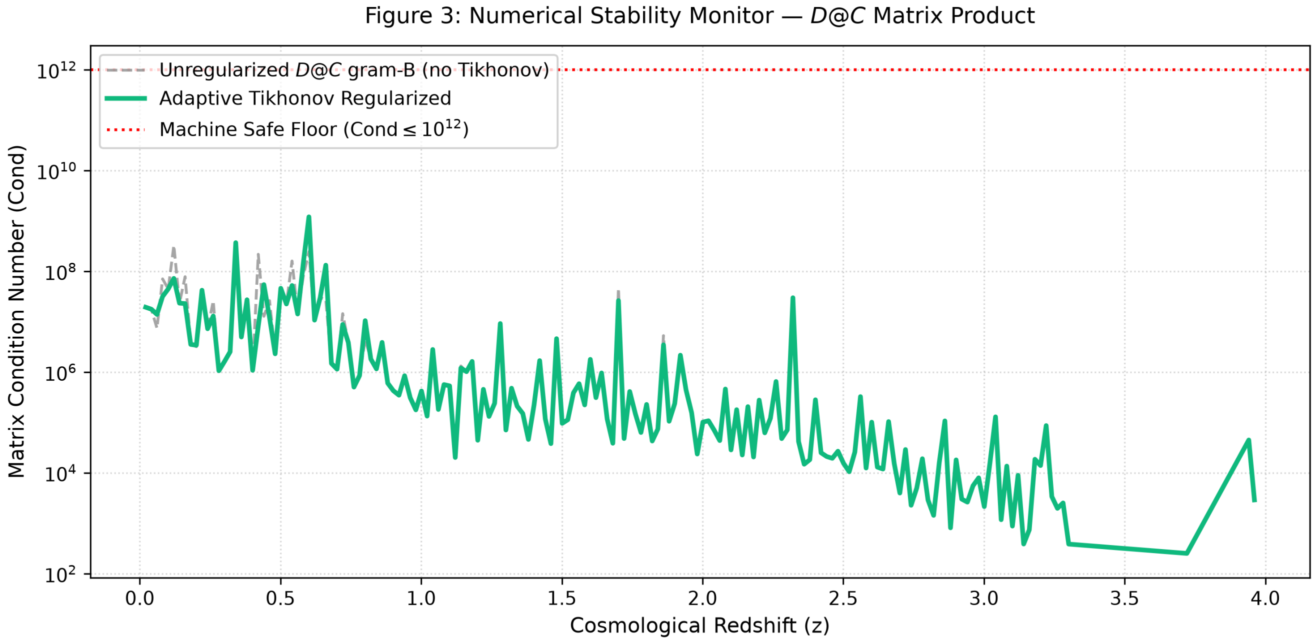

Figure 3 Numerical-stability monitoring of matrix-condition-number versus cosmological redshift z. Grey dashed curve: unregularised metric-matrix. Green solid curve: output under adaptive Tikhonov-manifold regularisation keeping condition-numbers inside numerically-stable domains. Red horizontal dotted line represents machine-safety ceiling $\text{Cond} \leq 10^{12}$.

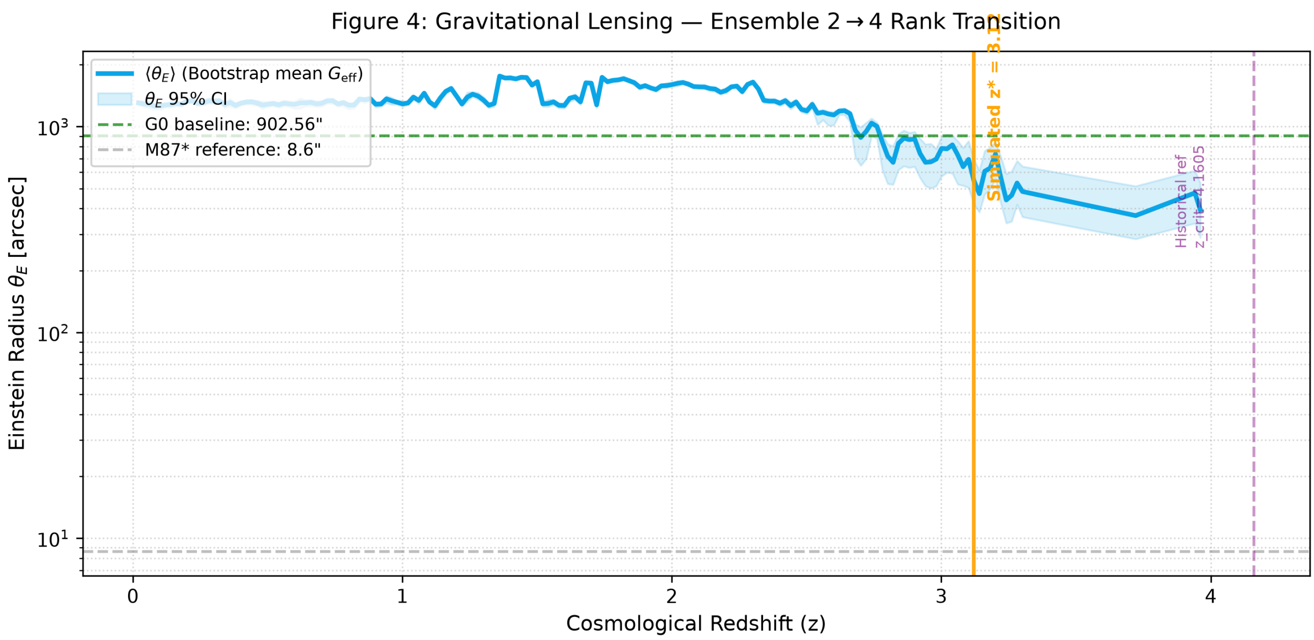

Figure 4 Gravitational-lensing Einstein-radius evolution with cosmic redshift. Blue solid curve: Bootstrap-ensemble-mean result; light-blue shaded band denotes 95 % confidence interval. Green dashed horizontal line: Einstein-radius baseline under G_0. Solid-orange vertical line marks simulated statistical transition $z^* = 3.13$; purple dashed vertical line marks historical-theoretical-reference $z_{\mathrm{crit}} = 4.1605$.

8 Discussion: Alignment-boundaries against SRE-v1.6 axiom-suite

Revision v6.2-rev preserves *all axioms* of the SRE-v1.6 framework without altering fundamental-principles; it only carries out mathematical-formulation upgrades, simulation-pipeline improvements and updates of numerical results:

1. **Emergent-ontology ultraviolet-boundary**: Planck-quantities are instance-realisation-cost thresholds, not fundamental network granularity.
2. **Dissipation-compensation duality**: distances are trace-inner-products of compensation- / dissipation-operators, mathematically realising v1.6 ontology of distance as book-keeping for topological-residual coherent-degradation.
3. **Mutual-measurement and Möbius light-residual**: photons correspond to Möbius-topology residual information-packets; BBP spectral-rank phase-transition splits single-handed Möbius topology into bidirectional two-layer networks.
4. **Uninstantiated-state / physical-rendering-layer**: eigenvalues crossing the Tracy-Widom boundary signify causal-interactions satisfying budget-constraints and completing instance-realisation.
5. **Homomorphic-mapping, not isomorphism**: astronomical-observations are coarse-grained many-to-one projections of the high-dimensional causal-network.
6. **Variable emergent-speed-of-light + conformal-covariance**: strictly preserves qualitative SRE-v1.6 predictions supplemented by explicit VSL formulae.
7. **Ontological-boundary statement**: this framework does not answer the ultimate origin of causal differences. It only describes how pre-existing asynchronous informational-differences give rise to emergent-cosmology. Questions concerning 0-to-1 genesis lie outside the closed scope of this revision.

> Version-historical reminder: original v6.1 draft contained residual coordinate-substrate ontological defects together with the a-priori conjecture $z_{\text{crit}} = 4.1605$. This v6.2-rev revision completes corrigenda. $z_{\text{crit}} = 4.1605$ serves only as historical-reference. $z^* = 3.13$ **is the statistical simulation output of this manuscript; it is not a direct astronomical observational measurement**. Versions v1.5.x and earlier are historical heuristic-sketches and shall be used for traceability purposes only.
> The Tracy-Widom rank-detector inside simulation-pipeline uses numerical-fitting implementation. Future work shall perform parameter-sensitivity-tests investigating the influence of sliding-window sizes and different observational catalogues upon z^*.

9 Future-research outlook

Subsequent work will couple the SRE-framework into CMB Boltzmann-solvers and test whether conformal-scaling of baryon-acoustic-horizon and photon-propagation during 4D-to-2D holographic-regression maintains CMB acoustic-peak positions within Planck / ACT observational error-bars. Simultaneously perform simulation-parameter-sensitivity-analysis, test the stability of simulated transition redshift z^* with DESI and other spectroscopic catalogues, and await future-telescope tests for the falsifiable imprints: gravitational-lensing 2-to-4 jump and chiral-polarisation signatures.

Conclusions

The SRE cosmic-gravity-framework (v6.2-rev) constructs a fully background-independent cosmological picture. Spacetime-topology, gravitational-strength and light-speed all emerge uniformly as macroscopic-consequences of link-topology within underlying discrete causal-information-networks. Using the SDSS/eBOSS spectroscopic dataset and 1500 Bootstrap Monte-Carlo realisations we obtain the BBP-spectral-rank statistical-phase-transition redshift $z^* = 3.13$. This phase-transition yields dimensional-crossover, gravitational-lensing factor-2-to-4 jump and primordial baryonic-cooling-enhancement effects which self-consistently alleviate the JWST high-redshift massive-galaxy puzzle. $z_{\text{crit}} = 4.1605$ is a historical-theoretical-reference originating from v6.1 and is no longer treated as a rigid prediction within this revision. The model delivers observationally-falsifiable imprints for examination by JWST and the Roman Space Telescope.